FYP PROJECT PROPOSAL

AI-Powered Desert Safari Booking & Camp Management Web Application

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Program: BS Information Technology

A centralized web platform for customer bookings, remote administration, AI customer support, package management, and driver coordination by phone or messages.

Prepared for Final Year Project Submission

1. Introduction

The project aims to develop a centralized web-based platform to manage a Desert Safari Camp located in Dubai, operated remotely from Pakistan. The system will automate customer bookings, payment records, package management, and camp operations.

The web application will allow customers to view packages and make bookings online, while the administrator can manage business operations through a centralized dashboard. Driver coordination will be handled through direct phone calls, WhatsApp messages, or other communication methods without developing a separate driver mobile application.

2. Problem Statement

Currently, desert safari bookings are managed manually through WhatsApp, Instagram, and phone calls. This can lead to double bookings, delayed responses, mismanagement, and a lack of proper record keeping.

A centralized web application is needed to manage bookings, customer information, payments, and camp operations efficiently while allowing the business to be managed remotely from Pakistan.

3. Objectives

- To create a unified web-based booking system for customers.
- To develop an admin panel for managing business operations remotely from Pakistan.
- To manage customer bookings, payments, packages, and camp capacity.
- To coordinate with drivers through phone calls, WhatsApp messages, and direct communication.
- To integrate an AI-powered chatbot for 24/7 customer support.
- To implement dynamic pricing based on demand, package availability, and other business rules.
- To provide route information and navigation links using mapping services.

4. System Modules

A. Customer Web Portal

Customers will be able to view desert safari packages, check availability, make bookings, and receive booking confirmations or e-tickets through supported communication channels.

B. Admin Panel

The administrator will have access to a centralized dashboard to manage bookings, customer records, payments, packages, camp capacity, and business reports. The system will support remote management from Pakistan.

C. Driver Coordination Module

No separate driver mobile application will be developed. The administrator will coordinate with drivers through direct phone calls, WhatsApp messages, or other communication methods. Driver assignments and relevant trip details can be recorded and managed through the web application's admin panel.

D. AI Engine

The AI engine will include a multilingual chatbot that answers common customer questions in English, Urdu, and Arabic; a dynamic-pricing component that recommends prices based on demand and availability; and route assistance through mapping services.

E. Booking and Payment Management

This module will maintain booking records, payment status, receipts, cancellations, and refunds according to administrator-defined policies.

F. Reports and Notifications

The system will generate booking summaries, revenue reports, package demand reports, and customer notifications through supported channels.

5. Technology Stack

- **Frontend:** React.js, HTML, CSS, JavaScript, and Bootstrap
- **Backend:** Node.js with Express
- **AI and Machine Learning:** Python and Scikit-learn
- **Database:** MongoDB or Firebase Firestore
- **AI Chatbot:** Dialogflow or Rasa
- **Maps and Routing:** Google Maps Platform API
- **Customer Messaging:** WhatsApp Business API, where required
- **Development Tools:** Visual Studio Code, Git, and GitHub

6. Expected Outcome

The expected outcome is a responsive and deployable web application that allows the Desert Safari business in Dubai to be managed remotely from Pakistan.

The system will centralize customer bookings, payments, package management, reporting, and AI-powered customer support. Driver coordination will remain flexible through direct phone calls and messages, eliminating the need to develop or maintain a separate driver application.

The system may later be expanded into a Software as a Service (SaaS) platform for other tourism companies.

7. Scope of the Project

- The project will focus on the customer web portal, administrator dashboard, booking management, database integration, AI chatbot, and basic dynamic pricing.
- Driver coordination will be handled manually through calls and messages, while the administrator records assignments and relevant trip details in the web application.
- The system will be designed to support future improvements and additional features as the business grows.